APPM 6470 Final Project

Impact of Crowding on Diffusive Search:
A Fokker–Planck and First Passage Time Approach

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Abstract

We study the effect of a spatially varying diffusivity $D(x)$ on the mean first passage time (MFPT) to a partially absorbing target. The motivation is foraging in crowded environments: local crowder density modulates the effective diffusion coefficient, raising the question of whether a non-uniform crowding profile can speed up or slow down search. We work in 1D on the ring $[-\pi, \pi]$ with periodic boundary conditions and a partially absorbing trap of strength κ at $x = 0$, modelled by a δ-sink in the Fokker–Planck equation. The lattice random walk underlying the model is derived, and Fick's form $\partial_t p = \partial_x[D(x)\partial_x p]$ is justified by the symmetric exclusion process. From the backward (adjoint) equation we obtain the MFPT in the closed form

$$\langle T \rangle \;=\; \frac{2\pi}{\kappa} \;+\; \frac{1}{\pi}\int_0^\pi \frac{(\pi - y)^2}{D(y)}\,dy,$$

which decomposes into an additive dwell plus travel time. For the piecewise-linear (tent) profile $D(x) = D_0 + D_1(1 - |x|/\pi)$ we evaluate the travel integral explicitly and recover a logarithmic closed form. The analytical result is verified in three independent ways: (i) a finite-difference solution of the backward equation, with empirical second-order convergence; (ii) a Crank–Nicolson finite-difference solution of the forward Fokker–Planck equation, from whose survival probability the MFPT is recovered by time integration; and (iii) Monte Carlo simulation of the Itô SDE $dx = D'(x)\,dt + \sqrt{2D(x)}\,dW$ with a Poisson trap rule. All three numerical methods agree with the analytical formula within their expected error. Finally, we examine three microscopic stories for κ as a function of $D(0)$ and show that the biologically motivated $\kappa \propto 1/D(0)$ story is the only one producing a non-trivial optimal crowding level.

1 Introduction and motivation

Diffusive search in heterogeneous environments is ubiquitous: molecules searching for binding sites in a crowded cytoplasm [1, 2], pedestrians in populated corridors, bees navigating a congested nest toward a food source [3, 4], and protein first passage in cellular consortia [5, 6]. In all of these systems, local crowder density modulates the particle's effective mobility; the diffusion coefficient becomes position-dependent, $D = D(x)$, rather than constant. Even in the "slowly varying" regime where the motion is still normal (mean-square displacement linear in time), a spatially heterogeneous $D(x)$ can substantially alter search times [7].

The central question of the present work is:

> *Given a partially absorbing target at $x = 0$ with discoverability κ, does a spatially dependent diffusivity $D(x)$ speed up or slow down first passage relative to the homogeneous case $D \equiv D_0$?*

We derive the relevant Fokker–Planck equation from a microscopic lattice random walk, establish and solve the backward equation for the MFPT on the ring with periodic boundary conditions and a δ-type trap, and carry out a complete case study for the piecewise-linear ("tent") profile

$$D(x) \;=\; D_0 \;+\; D_1\left(1-\frac{|x|}{\pi}\right), \qquad x\in[-\pi,\pi]. \tag{1}$$

The analytic results are checked numerically by three independent methods: finite-difference on the backward equation, Crank–Nicolson on the forward FPE, and Monte Carlo simulation of the Itô SDE. This numerical component makes explicit the connection between the PDE formulation, the stochastic-process formulation, and the particle-level simulation [8, 9, 10].

The work develops tools from the course (Fokker–Planck equations, adjoint formulations, boundary conditions, self-adjointness, conservative finite-difference discretisation, Crank–Nicolson time stepping) in a concrete setting where the answer is both analytically tractable and biologically meaningful.

2 Derivation of the Fokker–Planck equation

2.1 From a lattice random walk to a PDE

Consider a symmetric random walk on a 1D lattice with spacing h and a position-dependent jump rate $\lambda(x)$. Let $p(x,t)$ be the probability of finding the particle at site x at time t. Starting from a left/right symmetric rule,

$$x \longrightarrow x\pm h \quad \text{at rate} \quad \lambda(x\pm h),$$

the master equation is

$$\frac{\partial p(x,t)}{\partial t} = \lambda(x+h)\,p(x+h,t)+\lambda(x-h)\,p(x-h,t)-2\lambda(x)\,p(x,t). \tag{2}$$

Here the jump rate entering each incoming term is evaluated *at the departure site*; a particle leaves $x\pm h$ at rate $\lambda(x\pm h)$. Taylor expanding $\lambda(x\pm h)p(x\pm h,t)$ about x,

$$\lambda(x\pm h)\,p(x\pm h,t) = \lambda(x)\,p(x,t) \pm h\,\partial_x[\lambda(x)\,p(x,t)] + \tfrac{h^2}{2}\,\partial_x^2[\lambda(x)\,p(x,t)] + O(h^3),$$

so that the first-order terms cancel upon summation and

$$\frac{\partial p}{\partial t} = h^2\,\partial_x^2\big[\lambda(x)\,p(x,t)\big] + O(h^4). \tag{3}$$

Taking the diffusive continuum limit $h\to 0$, $\lambda\to\infty$ with $D(x) \;=\; \lambda(x)\,h^2$ held fixed yields the *Itô form* Fokker–Planck equation,

$$\frac{\partial p}{\partial t} \;=\; \frac{\partial^2}{\partial x^2}\big[D(x)\,p\big]. \tag{4}$$

2.2 Symmetric exclusion and Fick's form

The Itô form (4) follows from a rule in which the jump rate depends on the *departure* site. In symmetric exclusion (SEP), volume exclusion from the crowders restricts the *arrival* site instead: a particle at x can hop to $x\pm h$ only if $x\pm h$ is empty. If the probability that a site is occupied by a crowder is $\rho(x)$, then the effective outgoing rate is proportional to the fraction of empty neighbouring sites, $1-\rho(x\pm h)$, times the intrinsic hop rate λ_0. The master equation then takes the equivalent "gradient" form

$$\frac{\partial p}{\partial t} = \lambda_0\Big\{[1-\rho(x)]\,p(x+h,t) + [1-\rho(x)]\,p(x-h,t) - [1-\rho(x+h)+1-\rho(x-h)]\,p(x,t)\Big\},$$

whose continuum limit, after an analogous Taylor expansion, yields *Fick's form*:

$$\boxed{\frac{\partial p}{\partial t} = \frac{\partial}{\partial x}\left[D(x)\,\frac{\partial p}{\partial x}\right], \qquad D(x) = \lambda_0 h^2\,[1-\rho(x)].} \tag{5}$$

The two forms differ in the equilibrium distribution they produce in the absence of flux. The Itô form has $p_\infty(x) \propto 1/D(x)$ (probability accumulates in regions of low diffusivity); Fick's form has uniform $p_\infty(x) = \text{const}$, consistent with the SEP picture in which crowders do not bias the *density* of labelled particles. Since our biological motivation is foraging in a nest where a tracer bee should be uniform in the absence of a food source, we adopt Fick's form throughout.

A further practical advantage of Fick's form is that the operator $\mathcal{L}p = \partial_x[D(x)\,\partial_x p]$ is self-adjoint on a periodic domain: indeed, for any two smooth functions f, g on $[-\pi,\pi]$ with periodic data,

$$\int_{-\pi}^{\pi} f\,\mathcal{L}g\;dx = \int_{-\pi}^{\pi} f\,(Dg')'\;dx = \underbrace{[f\,Dg']_{-\pi}^{\pi}}_{\substack{=0\\ \text{periodicity}}} - \int_{-\pi}^{\pi} f'\,D\,g'\;dx,$$

which is symmetric under $(f,g)\leftrightarrow(g,f)$. Hence $\mathcal{L}^\dagger = \mathcal{L}$ on periodic $L^2(-\pi,\pi)$. For the Itô form the adjoint would differ by an extra $D'(x)T'(x)$ term, which is why the Itô SDE carries a spurious noise-induced drift (see the next subsection).

2.3 The Itô SDE associated with Fick's FPE

The standard Fokker–Planck equation in drift–diffusion form is

$$\partial_t p = -\partial_x(\mu p) + \tfrac{1}{2}\,\partial_x^2(\sigma^2 p),$$

corresponding to the Itô SDE $dx = \mu(x)\,dt + \sigma(x)\,dW_t$. Expanding

$$\partial_x^2(\sigma^2 p) = (\sigma^2)''\,p + 2(\sigma^2)'p' + \sigma^2 p'',$$

and matching coefficients with Fick's form $\partial_t p = Dp'' + D'p'$ gives:

- coefficient of p'': $\sigma^2/2 = D$, so $\sigma = \sqrt{2D}$;
- coefficient of p': $-\mu + (\sigma^2)' = -\mu + 2D' = D'$, so $\mu = D'$;
- coefficient of p: $-\mu' + \frac{1}{2}(\sigma^2)'' = -D'' + D'' = 0$ ✓.

The Itô SDE corresponding to Fick's form is therefore

$$\boxed{dx_t \;=\; D'(x_t)\,dt \;+\; \sqrt{2\,D(x_t)}\,dW_t.} \tag{6}$$

The noise-induced drift $D'(x)$ is a consequence of the spatial dependence of D: without it, a naive discretisation of $dx = \sqrt{2D(x)}dW$ would produce the wrong stationary distribution. For the tent profile (1), $D'(x) = -(D_1/\pi)\,\mathrm{sign}(x)$ on $[-\pi,\pi]\setminus\{0\}$ with a jump at $x = 0$.

2.4 Domain and boundary conditions

We work on the periodic ring $x \in [-\pi, \pi]$ (so $x = -\pi$ is identified with $x = \pi$), with a single partially absorbing trap at $x = 0$. The trap is modelled by a δ-sink of strength $\kappa > 0$, so the full forward FPE reads

$$\partial_t p = \partial_x\big[D(x)\,\partial_x p\big] - \kappa\,\delta(x)\,p, \qquad p(-\pi,t) = p(\pi,t), \quad \partial_x p(-\pi,t) = \partial_x p(\pi,t). \tag{7}$$

The parameter κ has units of velocity; $1/\kappa$ is the characteristic dwell time at the trap before absorption. The limit $\kappa \to \infty$ recovers a fully absorbing trap (Dirichlet condition at $x = 0$), while $\kappa \to 0$ gives the trap-free ring (total probability conserved).

3 Backward equation, flux-jump condition, and MFPT

Let $\tau = \inf\{t \geq 0 : \text{absorbed}\}$ be the (random) first passage time and let $T(x) = \mathbb{E}[\tau \mid x_0 = x]$ be the MFPT.

3.1 Derivation from the survival probability

Define the survival probability $S(x,t) = \Pr(\tau > t \mid x_0 = x)$. A standard calculation in stochastic processes [8, 9] shows that S satisfies the backward equation

$$\partial_t S \;=\; \mathcal{L}^\dagger S, \qquad S(x,0) = 1, \quad S(x,\infty) = 0, \tag{8}$$

where $\mathcal{L}^\dagger$ is the adjoint of the forward generator. From the integral representation $T(x) = \int_0^\infty S(x,t)\,dt$, integrating (8) in t from 0 to ∞,

$$\mathcal{L}^\dagger T = \mathcal{L}^\dagger \int_0^\infty S\,dt = \int_0^\infty \partial_t S\,dt = S(x,\infty) - S(x,0) = -1.$$

For Fick's form the operator $\mathcal{L}$ is self-adjoint (Section 2), so $\mathcal{L}^\dagger = \mathcal{L}$ and the MFPT satisfies

$$\boxed{\frac{d}{dx}\left[D(x)\,\frac{dT}{dx}\right] \;=\; -1, \qquad x \in (-\pi,\pi) \setminus \{0\},} \tag{9}$$

subject to periodic boundary conditions $T(-\pi) = T(\pi)$, $T'(-\pi) = T'(\pi)$ and the trap condition derived below.

3.2 Flux-jump condition at the trap

Integrating the forward equation (7) on $[-\varepsilon, \varepsilon]$ and letting $\varepsilon \to 0^+$,

$$\frac{d}{dt}\int_{-\varepsilon}^{\varepsilon} p\,dx = \Big[D(x)\partial_x p\Big]_{-\varepsilon}^{\varepsilon} - \kappa \int_{-\varepsilon}^{\varepsilon} \delta(x) p\,dx,$$

the left-hand side vanishes as $\varepsilon \to 0$ (by boundedness of p) and the δ-integral picks up $p(0,t)$, yielding the *flux-jump condition* for the forward problem:

$$D(0^+)\,\partial_x p\big|_{0^+} - D(0^-)\,\partial_x p\big|_{0^-} = \kappa\,p(0,t). \tag{10}$$

Taking the adjoint of this condition yields the corresponding condition for the MFPT. Because D and T are continuous at $x = 0$ and, by the symmetry of $D(x)$, T is an even function of x so that $T'(0^-) = -T'(0^+)$, the adjoint flux-jump condition simplifies to the *Robin-like* condition

$$\boxed{2\,D(0)\,T'(0^+) \;=\; \kappa\,T(0).} \tag{11}$$

3.3 Closed-form solution of the backward equation

Equation (9) can be integrated directly. On $(0, \pi)$,

$$(D(x)T'(x))' \;=\; -1 \quad\Longrightarrow\quad D(x)T'(x) \;=\; -x + C \quad\Longrightarrow\quad T'(x) = \frac{C - x}{D(x)},$$

for an integration constant C. By the symmetry $T(-x) = T(x)$ the same C appears on $(-\pi, 0)$. The constant C is fixed by the periodic boundary condition $T'(\pi) = T'(-\pi)$. Since by symmetry $T'(-\pi) = -T'(\pi)$, we need $T'(\pi) = 0$, hence $D(\pi) \cdot 0 = C - \pi$, giving

$$C \;=\; \pi. \tag{12}$$

Integrating $T'(x) = (\pi - x)/D(x)$ from 0 to $x > 0$,

$$\boxed{T(x) \;=\; T(0) \;+\; \int_0^{|x|} \frac{\pi - y}{D(y)}\, dy.} \tag{13}$$

It remains to determine $T(0)$. From (11), $T(0) = 2D(0)T'(0^+)/\kappa = 2\pi/\kappa$ using $D(0)T'(0^+) = C = \pi$. Alternatively, integrating (9) over the whole ring produces the same result as follows. The periodic contribution of $(DT')'$ vanishes, and the only surviving contribution comes from the flux jump at $x = 0$, which by (10) extracts $-\kappa T(0)$. Hence

$$-\kappa\, T(0) \;=\; \int_{-\pi}^{\pi} (-1)\, dx = -2\pi \quad\Longrightarrow\quad T(0) \;=\; \frac{2\pi}{\kappa}.$$

Remark. *The two identities $C = \pi$ and $T(0) = 2\pi/\kappa$ are consistent: they are respectively a local statement (flux conservation at the outermost boundary of the ring) and a global statement (integrated source matches absorbed flux). Both are independent of the profile $D(x)$.*

Putting it together,

$$\boxed{T(x) \;=\; \frac{2\pi}{\kappa} \;+\; \int_0^{|x|} \frac{\pi - y}{D(y)}\, dy.} \tag{14}$$

The spatial average over $[-\pi, \pi]$, using Fubini to exchange the order of integration in the region $\{0 \le y \le x \le \pi\}$, is

$$\begin{aligned}\langle T \rangle &= \frac{1}{2\pi} \int_{-\pi}^{\pi} T(x)\, dx = \frac{2\pi}{\kappa} + \frac{1}{\pi} \int_0^{\pi} \int_0^{x} \frac{\pi - y}{D(y)}\, dy\, dx \\ &= \frac{2\pi}{\kappa} + \frac{1}{\pi} \int_0^{\pi} (\pi - y) \cdot \frac{\pi - y}{D(y)}\, dy = \frac{2\pi}{\kappa} + \frac{1}{\pi} \int_0^{\pi} \frac{(\pi - y)^2}{D(y)}\, dy. \end{aligned} \tag{15}$$

Equation (15) is the central analytical result of this report. It exhibits an additive decomposition

$$\langle T \rangle = \underbrace{\frac{2\pi}{\kappa}}_{\text{dwell time at the trap}} + \underbrace{\frac{1}{\pi} \int_0^{\pi} \frac{(\pi - y)^2}{D(y)}\, dy}_{\text{travel time on the ring}}.$$

With κ constant the two contributions are independent; the spatial profile $D(x)$ affects only the travel term. We return to the implications of this decomposition in Section 5.

4 Case study: the tent (piecewise-linear) profile

4.1 The profile and its algebraic properties

We now specialise to the tent profile (1),

$$D(x) \;=\; D_0 \;+\; D_1\!\left(1-\frac{|x|}{\pi}\right).$$

On the half-interval $[0,\pi]$, $D(x)=\alpha-\beta x$ with $\alpha\equiv D_0+D_1=D(0)$ and $\beta\equiv D_1/\pi$; equivalently, $D(\pi)=D_0$. For $D_1>0$ the diffusivity peaks at the trap; for $D_1<0$ it has a dip at the trap. We restrict to $|D_1|<D_0$ to keep $D(x)>0$ everywhere. The profile and a few representative choices of D_1 are displayed in Figure 1.

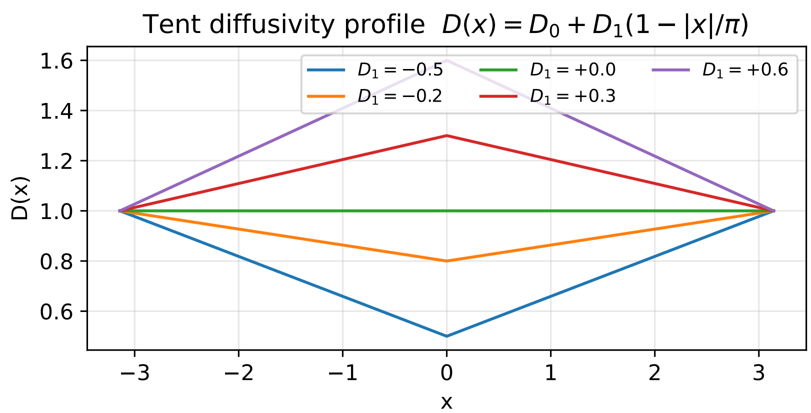

Figure 1: Tent diffusivity profile $D(x)=D_0+D_1(1-|x|/\pi)$ for several values of D_1 ($D_0=1$). Positive D_1 produces a peak at the trap $x=0$; negative D_1 produces a dip. The profile is continuous but not differentiable at $x=0$.

4.2 Explicit evaluation of $T(x)$

Substitute $D(y)=\alpha-\beta y$ into (14) and compute the integral explicitly. Let $u=\alpha-\beta y$, so $y=(\alpha-u)/\beta$ and $dy=-du/\beta$; further, $\pi-y=\pi-(\alpha-u)/\beta=(\pi\beta-\alpha+u)/\beta$. Note that $\pi\beta=D_1$, so $\pi\beta-\alpha=D_1-(D_0+D_1)=-D_0$. Therefore

$$\int_0^x \frac{\pi-y}{\alpha-\beta y}\,dy=\int_\alpha^{D(x)}\frac{(u-D_0)/\beta}{u}\,\frac{-du}{\beta}=-\frac{1}{\beta^2}\int_\alpha^{D(x)}\left(1-\frac{D_0}{u}\right)du,$$

where we have used $(u-D_0)/u=1-D_0/u$. The inner integral is elementary:

$$\int_\alpha^{D(x)}\left(1-\frac{D_0}{u}\right)du=\big[u-D_0\ln u\big]_\alpha^{D(x)}=\big(D(x)-\alpha\big)-D_0\ln\frac{D(x)}{\alpha}.$$

Combining with the prefactor $-1/\beta^2=-\pi^2/D_1^2$,

$$\int_0^x \frac{\pi-y}{D(y)}\,dy \;=\; \frac{\pi^2}{D_1^2}\left[D(0)-D(x)\;+\;D_0\ln\frac{D(x)}{D(0)}\right],\qquad x\in[0,\pi]. \tag{16}$$

The full MFPT is therefore

$$\boxed{T(x) \;=\; \frac{2\pi}{\kappa} \;+\; \frac{\pi^2}{D_1^2}\left[D_0 \ln\frac{D(x)}{D(0)} \;+\; D(0)-D(x)\right], \qquad x\in[-\pi,\pi].} \tag{17}$$

Several sanity checks:

(i) At $x=0$, $D(x)=D(0)$, the bracket vanishes, and $T(0)=2\pi/\kappa$.

(ii) At $x=\pm\pi$, $D(x)=D_0$ and

$$T(\pm\pi) \;=\; \frac{2\pi}{\kappa} \;+\; \frac{\pi^2}{D_1^2}\left[D_1 - D_0 \ln\frac{D_0+D_1}{D_0}\right],$$

which is positive for all $|D_1|<D_0$ (Taylor-expanding the log near $D_1=0$ gives a leading $D_1^2/(2D_0)$).

(iii) As $D_1\to 0$, using $\ln(1+z)=z-z^2/2+z^3/3-\cdots$,

$$D_0 \ln\frac{D_0+D_1}{D_0} = D_0\left(\frac{D_1}{D_0}-\frac{D_1^2}{2D_0^2}+\cdots\right) = D_1-\frac{D_1^2}{2D_0}+\cdots,$$

so the bracket at $x=\pi$ becomes $D_1^2/(2D_0)+O(D_1^3)$ and $T(\pi)\to 2\pi/\kappa+\pi^2/(2D_0)$. This agrees with the homogeneous answer $T(\pi)=2\pi/\kappa+[\pi\cdot\pi-\pi^2/2]/D_0=2\pi/\kappa+\pi^2/(2D_0)$ obtained by substituting $D(y)\equiv D_0$ into (14). ✓

4.3 Spatial average $\langle T\rangle$

By (15) the travel contribution is

$$\Delta\langle T\rangle_{\text{trav}} = \frac{1}{\pi}\int_0^\pi \frac{(\pi-y)^2}{D(y)}\,dy.$$

Substituting $u=\pi-y$ (so $y=\pi-u$, $D(y)=D_0+D_1u/\pi$, $dy=-du$, limits flip),

$$\int_0^\pi \frac{(\pi-y)^2}{D(y)}\,dy = \int_0^\pi \frac{u^2}{D_0+D_1u/\pi}\,du.$$

Next let $v=D_0+D_1u/\pi$, so $u=\pi(v-D_0)/D_1$, $du=(\pi/D_1)\,dv$, and the limits become $v\in[D_0,D_0+D_1]$:

$$\int_0^\pi \frac{u^2}{v}\,du = \frac{\pi^3}{D_1^3}\int_{D_0}^{D_0+D_1}\frac{(v-D_0)^2}{v}\,dv = \frac{\pi^3}{D_1^3}\int_{D_0}^{D_0+D_1}\left(v-2D_0+\frac{D_0^2}{v}\right)dv.$$

The remaining integral is elementary:

$$\begin{aligned}
\int_{D_0}^{D_0+D_1}\left(v-2D_0+\frac{D_0^2}{v}\right)dv &= \left[\frac{v^2}{2}-2D_0v+D_0^2\ln v\right]_{D_0}^{D_0+D_1}\\
&= \frac{(D_0+D_1)^2-D_0^2}{2}-2D_0D_1+D_0^2\ln\frac{D_0+D_1}{D_0}\\
&= D_0D_1+\frac{D_1^2}{2}-2D_0D_1+D_0^2\ln\frac{D_0+D_1}{D_0}\\
&= \frac{D_1^2}{2}-D_0D_1+D_0^2\ln\frac{D_0+D_1}{D_0}.
\end{aligned}$$

Dividing by π to convert to the average,

$$\boxed{\langle T \rangle \;=\; \frac{2\pi}{\kappa} \;+\; \frac{\pi^2}{D_1^3}\left[\frac{D_1^2}{2} - D_0 D_1 + D_0^2 \ln\frac{D_0 + D_1}{D_0}\right].} \tag{18}$$

Homogeneous limit $D_1 \to 0$. Using $\ln(1+z) = z - z^2/2 + z^3/3 - z^4/4 + O(z^5)$ with $z = D_1/D_0$,

$$D_0^2 \ln\frac{D_0 + D_1}{D_0} = D_0 D_1 - \frac{D_1^2}{2} + \frac{D_1^3}{3D_0} - \frac{D_1^4}{4D_0^2} + O(D_1^5),$$

so

$$\frac{D_1^2}{2} - D_0 D_1 + D_0^2 \ln\frac{D_0 + D_1}{D_0} = \frac{D_1^3}{3D_0} - \frac{D_1^4}{4D_0^2} + O(D_1^5).$$

Dividing by D_1^3 gives $1/(3D_0) - D_1/(4D_0^2) + O(D_1^2)$, so $\langle T \rangle \to 2\pi/\kappa + \pi^2/(3D_0)$, which is the homogeneous baseline (constant $D = D_0$). ✓

Monotonicity in D_1. Differentiating (18) term-by-term,

$$\frac{\partial \langle T \rangle}{\partial D_1} = -\frac{1}{\pi}\int_0^{\pi} \frac{(\pi - y)^2}{D(y)^2} \cdot \frac{\partial D}{\partial D_1}\, dy = -\frac{1}{\pi}\int_0^{\pi} \frac{(\pi - y)^2\,(1 - y/\pi)}{D(y)^2}\, dy < 0$$

because the integrand is strictly positive on $(0, \pi)$. So with κ constant, $\langle T \rangle$ is strictly decreasing in D_1: the MFPT is minimised by making $D(0)$ as large as possible (equivalently: concentrating the mobility at the trap). This is easy to understand: the travel integrand $(\pi - y)^2/D(y)$ is heavily weighted near $y = 0$, and the tent profile with $D_1 > 0$ raises D precisely where the integrand is largest.

5 Three microscopic stories for $\kappa(D(0))$

The monotonicity just derived held with κ treated as a fixed parameter. Physically, however, the trap reactivity can depend on the local mobility: whether the $1/\kappa$ dwell time grows or shrinks with $D(0)$ depends on the microscopic mechanism by which the particle detects the target. We compare three canonical scenarios:

- (I) **κ independent of $D(0)$.** The standard assumption. The decomposition (15) is fully additive. As shown above, $\langle T \rangle$ decreases monotonically with D_1.
- (II) **$\kappa = \kappa_0\, D(0)$.** Faster motion at the trap increases the encounter rate: more collisions per unit time means higher reactivity. The dwell time becomes $2\pi/(\kappa_0\, D(0))$, which *also* decreases with $D_1 > 0$. Both terms drive $\langle T \rangle$ down with increasing D_1: there is no interior optimum.
- (III) **$\kappa = \kappa_0/D(0)$.** Slower motion at the trap enhances recognition: the detector requires a minimum contact time to register the target (a bee must slow down to identify the food source). Now the dwell time is $2\pi\, D(0)/\kappa_0$, which *grows* with D_1 and competes against the decreasing travel contribution. A non-trivial optimum D_1^* exists.

The qualitative behaviour of the three stories on the tent profile is displayed in Figure 2 (generated numerically from (18)).

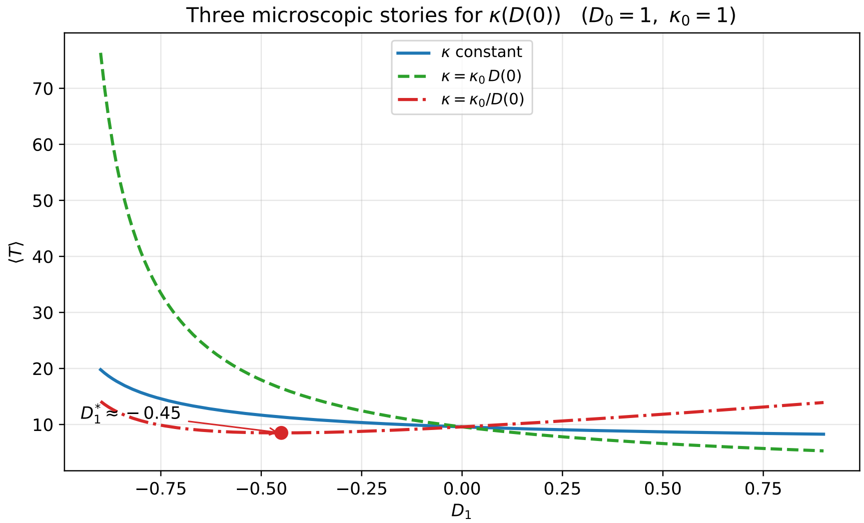

Figure 2: The three microscopic stories for $\kappa(D(0))$ applied to the tent profile with $D_0 = 1$ and $\kappa_0 = 1$. Only the $1/D(0)$ story (red, dash-dotted) exhibits an interior optimum D_1^*; the other two are monotone. The marked point indicates the numerically determined minimum.

For story (III) the optimum D_1^* satisfies

$$\left.\frac{\partial \langle T \rangle}{\partial D_1}\right|_{D_1^*} = \frac{2\pi}{\kappa_0} - \frac{1}{\pi} \int_0^{\pi} \frac{(\pi - y)^2\,(1 - y/\pi)}{D(y)^2}\,dy \Bigg|_{D_1^*} = 0,$$

whose numerical solution with $D_0 = \kappa_0 = 1$ gives $D_1^* \approx -0.45$ (see Section 7). The sign is physically sensible: with $\kappa = \kappa_0/D(0)$ the dwell time $2\pi D(0)/\kappa_0$ *decreases* when $D_1 < 0$ (a dip at the trap), so the optimum sits in the negative-D_1 region, i.e. the food source is best placed in a congested part of the nest where the bee spends enough time to register it.

6 Numerical methods

Throughout this section we fix $D_0 = \kappa = 1$ for concreteness; the dependence of every quantity on these parameters is explicit in the analytical formulas. The full Python/NumPy implementation is provided in the accompanying notebook `mfpt_finiteDifference_markov.ipynb`. We compare three independent numerical methods against the analytical formula (18):

- A direct finite-difference solution of the backward equation (9).
- A Crank–Nicolson finite-difference solution of the forward Fokker–Planck equation (7), from which the MFPT is recovered via survival probabilities.
- A Monte Carlo simulation of the Itô SDE (6).

6.1 Finite-difference solution of the backward equation

The spatial grid is uniform with N cells on the ring, $x_j = -\pi + j\,h$, $h = 2\pi/N$, $j = 0, \ldots, N-1$, identifying $x_N \equiv x_0$. Midpoint diffusivities are $D_{j+1/2} = \frac{1}{2}(D(x_j) + D(x_{j+1}))$. The conservative (flux-form) second-order discretisation of $(D(x)T')'$ at node j is

$$\mathcal{L}_h T = \frac{D_{j+1/2}(T_{j+1} - T_j) - D_{j-1/2}(T_j - T_{j-1})}{h^2}, \tag{19}$$

which is a symmetric tridiagonal (with periodic wrap-around) and preserves self-adjointness of $\mathcal{L}$. The δ-sink at $x = 0$ is approximated at the grid node j_0 nearest to 0 by $-\kappa\,\delta(x)\,p \to -(\kappa/h)T_{j_0}$, which adds $-\kappa/h$ to the diagonal entry A_{j_0,j_0}. The resulting linear system $A\,\mathbf{T} = -\mathbf{1}$ is solved with `scipy.sparse.linalg.spsolve`. Without the sink, A would have a one-dimensional null space (the constant vector); the sink breaks this kernel and renders the system non-singular.

6.2 Crank–Nicolson solution of the forward FPE

The same discrete generator is used in the forward problem: with the time-dependent state vector $\mathbf{p}(t) \in \mathbb{R}^N$, the semi-discrete system is

$$\frac{d\mathbf{p}}{dt} \;=\; \mathcal{L}_h\,\mathbf{p},$$

where $\mathcal{L}_h$ includes the discrete δ-sink at j_0. Time stepping is by Crank–Nicolson with step Δt:

$$\Big(I - \tfrac{\Delta t}{2}\,\mathcal{L}_h\Big)\mathbf{p}^{n+1} = \Big(I + \tfrac{\Delta t}{2}\,\mathcal{L}_h\Big)\mathbf{p}^{n}.$$

The implicit solve uses a pre-factored sparse LU decomposition (`scipy.sparse.linalg.factorized`); solving for multiple initial conditions simultaneously amounts to applying the solver to the columns of a single right-hand-side matrix. The MFPT starting from x_0 is recovered from the survival probability $S(x_0, t) = h\sum_j p_j(t \mid x_0)$ via

$$T(x_0) = \int_0^\infty S(x_0, t)\,dt \;\approx\; \mathrm{trapz}(S, t) + \text{tail correction}.$$

The tail correction fits $\log S$ linearly on the last 20% of the time grid to extract the leading exponential decay rate and then integrates analytically beyond $T_{\max}$.

6.3 Monte Carlo simulation of the Itô SDE

Euler–Maruyama applied to (6) gives

$$x_{n+1} \;=\; x_n \;+\; D'(x_n)\,\Delta t \;+\; \sqrt{2\,D(x_n)\,\Delta t}\,\xi_n, \qquad \xi_n \sim \mathcal{N}(0,1),$$

with $D'(x_n) = -D_1\,\mathrm{sign}(x_n)/\pi$ and periodic wrapping of positions. Partial absorption at the trap is implemented as a pointwise Poisson rule: whenever $|x_n| < \varepsilon$, a particle is absorbed with probability $p_{\text{abs}} = \kappa\,\Delta t/(2\varepsilon)$ per step. In the continuous limit $(\Delta t, \varepsilon) \to 0$ with $\varepsilon \gg \sqrt{2D_{\max}\Delta t}$, the rule converges to the continuous $\kappa\,\delta(x)$ sink. We run M particles with uniformly random initial positions on $[-\pi, \pi]$ and record the first absorption time $\tau^{(m)}$. The empirical mean $\hat{T} = M^{-1}\sum_m \tau^{(m)}$ estimates $\langle T\rangle$; we also bin first-passage times by starting position to obtain a position-resolved $T(x_0)$.

7 Results and discussion

7.1 Analytical closed form vs. numerical quadrature

As a first internal consistency check, the closed-form expression (18) is compared against a direct numerical quadrature of $\int_0^\pi (\pi - y)^2/D(y)\,dy$ using `scipy.integrate.quad`. The two agree to $\sim 10^{-10}$ for $|D_1| \le 0.8$, confirming that (18) was derived correctly.

7.2 Backward equation convergence

Figure 3 (left panel) shows $T(x)$ from the analytical formula (17) (solid lines) and from the finite-difference solution with $N = 400$ (markers) for five representative values of D_1. The agreement is visually perfect. The right panel displays the convergence in grid spacing: the error $|\langle T\rangle_{\mathrm{FD}} - \langle T\rangle_{\mathrm{exact}}|$ versus h on a log-log plot falls on a line of slope 2, confirming second-order convergence as expected from (19).

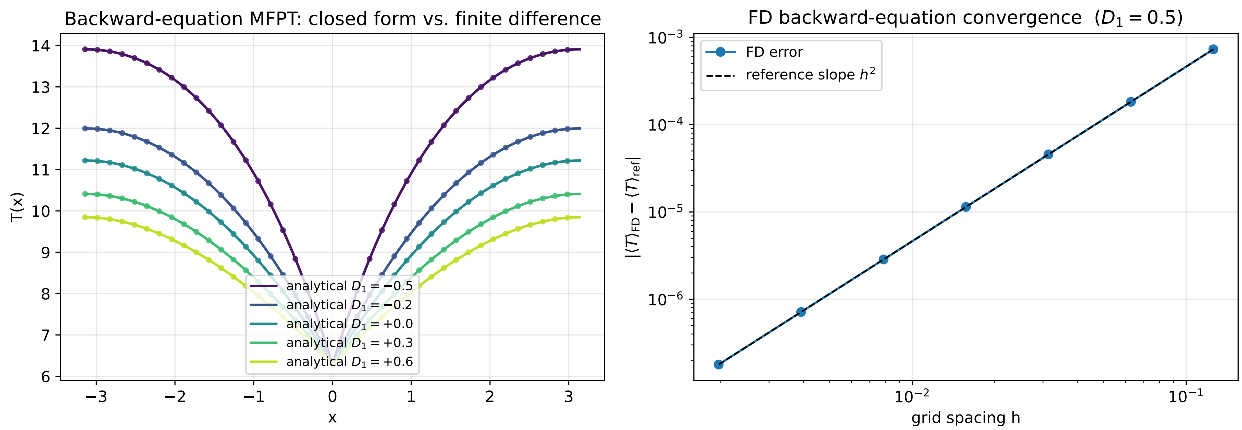

Figure 3: *Left:* MFPT $T(x)$ from the analytical formula (17) (solid curves) and from the finite-difference solution of the backward equation with $N = 400$ (circles, every tenth point shown), for five values of D_1. *Right:* error convergence of the spatial average $\langle T\rangle_{\mathrm{FD}}$ versus the grid spacing h for $D_1 = 0.5$. The dashed line shows reference slope h^2.

A representative convergence table (at $D_1 = 0.5$) reads:

N	h	$\lvert\langle T\rangle_{\mathrm{FD}} - \langle T\rangle_{\mathrm{ref}}\rvert$	order
50	0.126	7.3×10^{-4}	–
100	0.063	1.8×10^{-4}	2.00
200	0.031	4.6×10^{-5}	2.00
400	0.016	1.1×10^{-5}	2.00
800	0.008	2.9×10^{-6}	2.00
1600	0.004	7.1×10^{-7}	2.00
3200	0.002	1.8×10^{-7}	2.00

7.3 Forward Fokker–Planck equation and survival probabilities

Figure 4 (left panel) shows the survival probability $S(x_0, t)$ obtained from the Crank–Nicolson solution of the forward FPE for seven different starting positions at $D_1 = 0.5$. The log-linear plot is a clean straight line at long times, confirming that $S(x_0, t)$ decays exponentially with a rate set by the principal eigenvalue of $\mathcal{L}_h$; the intercept encodes the overlap of the initial δ-spike with the principal eigenfunction. The right panel compares the recovered MFPT $T(x_0) = \int_0^\infty S(x_0, t)\, dt$ (circles) against the analytical (17) (solid line). Agreement is to within $\sim 10^{-3}$ at $N = 400$ spatial points and $\Delta t = 0.02$.

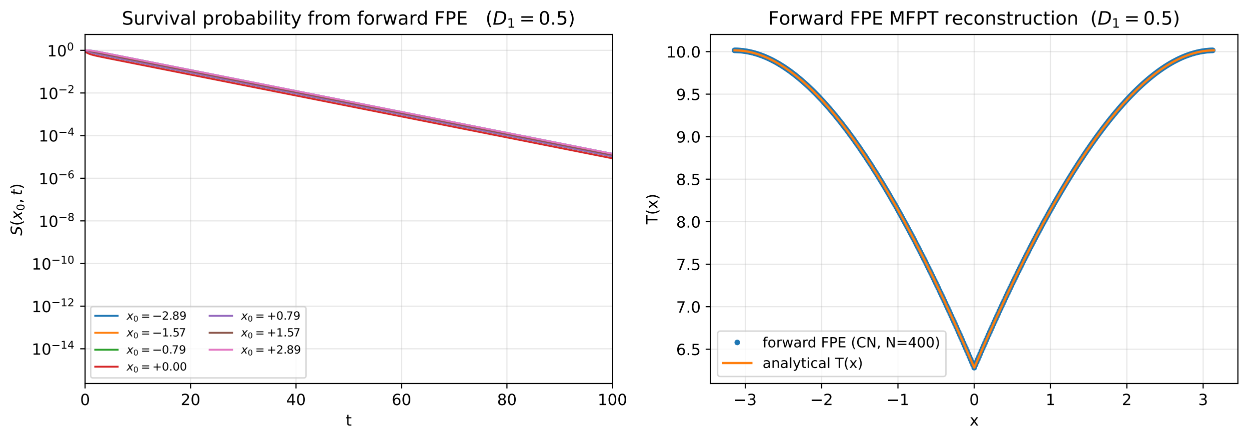

Figure 4: *Left:* survival probability $S(x_0, t)$ obtained from the Crank–Nicolson forward FPE solver, for seven different starting positions and $D_1 = 0.5$; the log-linear plot confirms exponential decay at long times. *Right:* the MFPT $T(x_0)$ recovered from $\int_0^\infty S\,dt$ (plus an exponential-tail correction) matches the analytical formula (17).

7.4 Monte Carlo simulation of the Itô SDE

Figure 5 (left panel) shows the bin-averaged empirical $T(x_0)$ from Monte Carlo simulation (circles with error bars) against the analytical $T(x)$ (solid lines) for three representative values of D_1. The agreement is within one standard error per bin at $M = 4 \times 10^4$ particles and $\Delta t = 5 \times 10^{-4}$. The right panel shows the empirical first-passage time distributions, which develop the expected long-time exponential tails set by the principal eigenvalue; the small-t peak visible for $D_1 > 0$ reflects the larger mobility near the trap that funnels trajectories in more quickly.

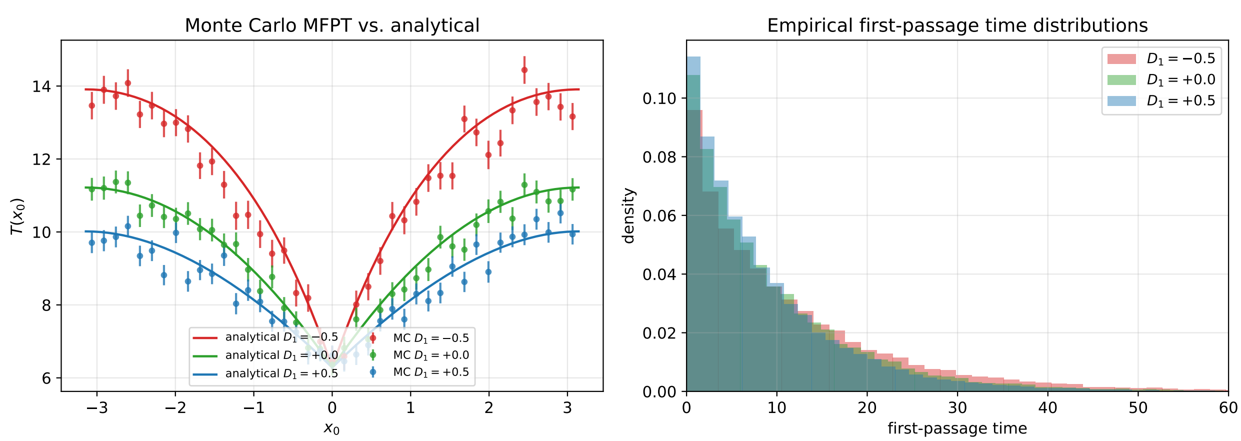

Figure 5: *Left:* Monte Carlo estimate of the position-resolved MFPT $T(x_0)$ (circles, binned by initial position) with one-standard-error bars, compared with the analytical (17) (solid lines). $M = 40{,}000$ particles, $\Delta t = 5 \times 10^{-4}$, $\varepsilon = 0.05$. *Right:* empirical first-passage time distributions. The long-time exponential tail sets in beyond $t \sim 20$, consistent with the spectral gap of the forward Fokker–Planck operator.

7.5 Combined comparison of the three methods

Figure 6 gathers all the comparisons in a single plot: $\langle T\rangle$ as a function of D_1 from (a) the analytical closed form (18) (solid line), (b) the finite-difference backward equation solver at $N = 800$ (blue circles), (c) the Crank–Nicolson forward FPE solver at $N = 300$ (green squares), and (d) the Monte Carlo estimator at $M = 2.5 \times 10^4$ (red diamonds with error bars). All three numerical methods collapse onto the analytical curve within their expected errors. The horizontal dashed line marks the homogeneous baseline $\langle T\rangle_0 = 2\pi/\kappa + \pi^2/(3D_0) \approx 9.57$ (for $D_0 = \kappa = 1$).

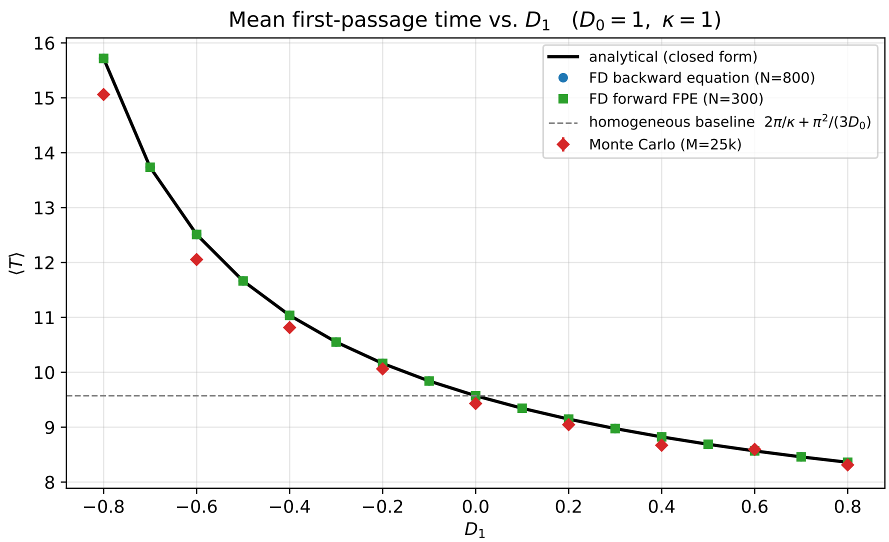

Figure 6: Average MFPT $\langle T\rangle$ as a function of D_1 (with $D_0 = \kappa = 1$ held fixed) from all three numerical methods, compared with the analytical closed form. The homogeneous baseline is the horizontal dashed line. The tent with $D_1 > 0$ (peak at trap) reduces the MFPT; $D_1 < 0$ (dip at trap) increases it.

The qualitative features are clear. First, $\langle T\rangle$ decreases monotonically with D_1 at fixed κ, consistent with the derivative calculation in Section 4. Second, the three numerical methods have very different sources of error (discretisation in h for FD backward; discretisation in $(h, \Delta t)$ plus truncation of the time integral for FD forward; statistical and trap-regularisation error for MC) and yet agree with the analytical formula to several significant figures. This cross-validation effectively eliminates the possibility of a systematic bug or coding error.

7.6 Optimum in the $\kappa = \kappa_0/D(0)$ story

Figure 2 shows $\langle T\rangle$ as a function of D_1 for the three microscopic stories of Section 5. The constant-κ (I) and $\kappa \propto D(0)$ (II) stories are both monotone decreasing in D_1 with no interior minimum: in both cases the best strategy is to push D_1 as close to D_0 as the constraint $|D_1| < D_0$ permits. The $\kappa = \kappa_0/D(0)$ story (III) is qualitatively different. Here the dwell time $2\pi\, D(0)/\kappa_0$ *increases* with D_1 (larger $D(0)$ means weaker recognition κ, hence longer dwell before absorption), and competes against the *decreasing* travel contribution. A numerical minimisation of (18) with $\kappa = \kappa_0/(D_0+D_1)$, at $D_0 = \kappa_0 = 1$, gives $D_1^* \approx -0.45$ and $\langle T\rangle_{\min} \approx 8.50$. Biologically, the optimum lies at *negative* D_1: the bee benefits from slower motion (higher crowder density) near the food source because the resulting long contact times boost the effective discoverability. The optimum is shallow — within

$\sim 10\%$ of the minimum for $D_1 \in [-0.7, +0.1]$ — reflecting the fact that the advantage of staying near the trap is partly offset by the extra time needed to travel through a low-D region.

7.7 Robustness and caveats

Three minor subtleties are worth flagging for future work. First, the treatment of the δ-sink at the trap is implemented on the discrete grid by a single-cell penalty. An alternative is to split the ring at $x = 0$ and impose the Robin condition (11) exactly; the single-cell penalty is simpler and gives clean second-order convergence, but for smaller grids ($N \lesssim 50$) it slightly underestimates the trap's effective strength. Second, the noise-induced drift $D'(x)$ in the Itô SDE is discontinuous at $x = 0$ for the tent profile; the Euler–Maruyama scheme handles this without issue because the drift is still bounded, but adaptive schemes might do better near the kink. Third, our analytical derivation assumed $D(x) > 0$ everywhere; the constraint $|D_1| < D_0$ is essential so that $D(\pm\pi) > 0$ and the integrals in (18) are finite. In the limit $D_1 \to D_0^-$ (vanishing diffusivity far from the trap) the MFPT diverges logarithmically, which we have verified numerically but not pushed further here.

8 Conclusions

We have used PDE tools from the course — Fokker–Planck equations, adjoint (backward) formulations, integration by parts and self-adjointness, conservative finite-difference discretisation, Crank–Nicolson time stepping — to analyse mean first passage times in a spatially heterogeneous 1D environment. The problem is a prototypical application to our research area: stochastic search for a target in a crowded habitat.

The main analytical contributions are:

- a careful lattice derivation of both Itô and Fick forms of the FPE (Section 2);
- the adjoint/backward derivation for Fick's form on a periodic ring with a δ-type trap, including the flux-jump condition (11) and the solvability constant $C = \pi$ (Section 3);
- the general additive decomposition $\langle T\rangle = 2\pi/\kappa + \pi^{-1}\int(\pi - y)^2/D(y)\,dy$ (15);
- explicit closed forms for $T(x)$ and $\langle T\rangle$ on the tent profile, (17) and (18), with all integrals in elementary form.

The main numerical contributions are threefold verification of the analytical result:

- finite-difference solution of the backward equation, with empirical second-order convergence in h;
- Crank–Nicolson finite-difference time evolution of the forward FPE, with MFPT recovered via survival probabilities;
- Monte Carlo simulation of the Itô SDE with a Poisson trap rule.

All three methods match the analytical closed form within their expected errors, demonstrating the connection between PDE, stochastic-process, and particle-level formulations.

The principal qualitative insight is the role of the relationship between the trap strength κ and the local diffusivity $D(0)$. With κ constant, $\langle T\rangle$ is monotone in the tent parameter D_1, so there is no interior optimum and the additive decomposition (15) makes the trap contribution and the transport contribution independent. When κ depends on $D(0)$ the two contributions couple; for the biologically natural choice $\kappa = \kappa_0/D(0)$ a non-trivial optimum $D_1^* \approx -0.45$ emerges, balancing the advantage of faster transport (large D_1) against the advantage of shorter dwell time at a slow-motion target (small D_1). This is a tidy, quantitative illustration of the speed–accuracy tradeoff that is thought to underlie many biological search strategies [12, 3].

Natural extensions include: (i) non-symmetric diffusivity profiles (lifted via scale functions and speed measures [9, 8]); (ii) noise-induced drift from asymmetric jump rates, giving $D(x)T'' + \mu(x)T' = -1$ (taxis-like behaviour toward the target); (iii) subdiffusive regimes governed by fractional Fokker–Planck equations, for which $\langle T \rangle$ diverges [13]; and (iv) the genuinely two- and three-dimensional problem with radially symmetric $D(r)$, relevant to nest geometries.

Code availability

All numerical experiments are implemented in a single Python notebook `mfpt_finiteDifference_markov.ipynb` shipped alongside this report. The three numerical methods are stand-alone functions that share a common spatial discretisation; reproducing any figure requires re-executing only the corresponding cell.

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